

Linkcoin

The Linking Layer for the UTXO Economy

A Blockchain Infrastructure for Connecting Value, Data, and Identity

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Linkcoin Development Team

Abstract

Linkcoin began as an experimental fork exploring the boundaries of UTXO-based cryptocurrencies. Through focused development, it has evolved into a specialized blockchain designed to *link*—connecting transactions atomically, bridging separate blockchains trustlessly, anchoring data immutably, and enabling scalable multi-party payments. This whitepaper introduces LinkNet, a comprehensive protocol layer built on Linkcoin’s unique IronTX foundation, positioning Linkcoin as essential infrastructure for the interconnected blockchain economy.

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1 Introduction

1.1 From Experiment to Infrastructure

Linkcoin’s journey began as many cryptocurrency projects do—as a fork exploring new ideas. The original Linkcoin was a lightweight experiment, a “memecoin” in the truest sense: a playground for learning and community building.

But within that experimentation emerged a vision: **what if a blockchain’s primary purpose was to link things together?**

Not just linking inputs to outputs in transactions, but:

- Linking UTXOs with enforceable relationships
- Linking blockchains through trustless atomic swaps
- Linking real-world data to immutable on-chain proofs
- Linking payment channel participants beyond two parties

This whitepaper presents Linkcoin 2.0: a purpose-built linking infrastructure for the UTXO economy.

1.2 Design Philosophy

Linkcoin adheres to five core principles:

1. **UTXO Purity:** Maintain Bitcoin’s proven security model
2. **Minimal Trust:** Eliminate intermediaries wherever possible
3. **Off-Chain Scaling:** Reserve the blockchain for settlement
4. **Interoperability:** Connect with the broader blockchain ecosystem
5. **Simplicity:** Achieve complex outcomes with simple primitives

1.3 Key Innovations

Innovation	Description
IronTX	Non-malleable transactions for safe pre-signed contracts
Embedded Lightning	Native payment channels without external dependencies
UTXO Links	Programmable relationships between transaction outputs
Cross-Chain Links	Trustless atomic swaps via adaptor signatures
Data Links	On-chain anchoring for identity and attestations

Table 1: Linkcoin Key Innovations

2 The Problem: Disconnected Blockchains

2.1 The Isolation Problem

Despite a decade of blockchain development, the ecosystem remains fragmented. Bitcoin, Ethereum, Litecoin, and other chains exist as isolated islands with no native connection between them.

Solution	Trust Model	Failure Mode
Centralized Exchanges	Full custody	Exit scams, hacks (\$3B+ lost)
Federated Bridges	Multi-sig committee	Collusion, key compromise
Wrapped Tokens	Smart contract	Bugs, oracle manipulation
Hash Time-Locked Contracts	Minimal	Free option problem

Table 2: Cross-Chain Solution Comparison

2.2 Current Solutions and Their Flaws

2.3 The Malleability Barrier

Bitcoin’s original transaction malleability prevented reliable pre-signed transactions. When a transaction’s ID can change after signing, child transactions that reference it become invalid. This blocked:

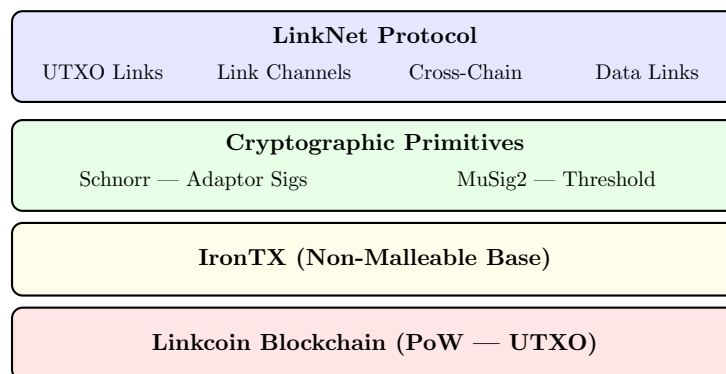
- Safe Lightning channel construction
- Atomic swap reliability
- Multi-party contract protocols

Linkcoin’s IronTX provides a complete solution.

3 The Linkcoin Solution

3.1 Architecture Overview

Linkcoin introduces LinkNet: a unified protocol layer for linking value, data, and identity.



3.2 The Linking Thesis

Every valuable action in the blockchain economy involves *linking*:

Action	What’s Being Linked
Payment	Sender → Receiver
Smart Contract	Conditions → Outcomes
Identity	Person → Public Key
Atomic Swap	Chain A TX → Chain B TX

Table 3: The Linking Paradigm

Linkcoin makes linking a first-class primitive.

4 Technical Foundation: IronTX

4.1 The Malleability Fix

IronTX introduces canonical transaction serialization, eliminating all malleability vectors. Every transaction has exactly one valid encoding.

4.2 IronTxId: Stable Transaction Identifiers

IronTX introduces a secondary transaction identifier excluding signature data:

$$\text{Standard TxId} = \text{Hash}(\text{version}||\text{inputs}||\text{outputs}||\text{locktime}||\text{signatures})$$

$$\text{IronTxId} = \text{Hash}(\text{version}||\text{inputs}||\text{outputs}||\text{locktime})$$

The IronTxId remains **stable** regardless of signature variations.

4.3 Activation Status

Network	IronTX Height	Status
Mainnet	Block 12,000	Active
Testnet	Block 12,000	Active
Regtest	Block 0	Active

Table 4: IronTX Activation

4.4 Schnorr Signatures

Linkcoin implements BIP-340 Schnorr signatures natively:

- **Linearity:** Enables signature aggregation (MuSig2)
- **Batch Verification:** 2-3x faster block validation
- **Adaptor Signatures:** Building block for atomic operations
- **Simpler Security:** Provable in random oracle model

5 LinkNet Protocol Suite

5.1 UTXO Links

UTXO Links create enforceable relationships between transaction outputs.

5.1.1 Link Types

Atomic Links Multiple UTXOs must be spent in the same transaction

Sequence Links UTXO B unlocks only after UTXO A is spent

Provenance Links Cryptographic chain tracking asset history

5.2 Link Channels

Link Channels extend payment channels beyond two parties.

5.2.1 Channel Factories

Open multiple channels with a single on-chain transaction:

- 3x reduction in on-chain transactions
- Shared liquidity pool
- Atomic multi-channel operations

5.2.2 Virtual Channels

Direct channels through intermediaries without trust. Using adaptor signatures, the virtual channel is cryptographically secured.

5.2.3 Linked Payments

Atomic multi-destination payments: either all complete or none complete.

5.3 Cross-Chain Links

Trustless connections to other UTXO blockchains via adaptor signature atomic swaps.

Adaptor Signature Atomic Swap

1. Alice generates secret s , publishes $T = s \cdot G$
2. Both create adaptor signatures locked to T
3. Alice reveals s by claiming on Chain B
4. Bob extracts s from published signature, claims on Chain A

Result: Trustless, atomic, no intermediary

5.3.1 Supported Chains

Chain	Signature	Support
Bitcoin (Taproot)	Schnorr	Native
Litecoin (MWEB)	Schnorr	Native
Bitcoin (SegWit)	ECDSA	Via 2P-ECDSA
Ethereum	ECDSA	Via 2P-ECDSA

Table 5: Cross-Chain Compatibility

5.4 Data Links

On-chain anchoring for identity and attestations.

5.4.1 Document Notarization

Embed document hash in transaction for immutable timestamp proof.

5.4.2 Decentralized Identity

Linkcoin implements a native DID method: `did:link:<address>`

5.4.3 Oracle System

Oracles commit to future data revelations, enabling trustless conditional payments.

6 Embedded Lightning Network

Unlike Bitcoin where Lightning requires external software, Linkcoin embeds Lightning directly in the core:

Aspect	External LN	Embedded LN
Setup complexity	High	None
Peer network	Separate	Unified P2P
Key management	Separate keys	Unified keychain
User experience	Technical	Mass adoption ready

Table 6: Lightning Implementation Comparison

7 Economic Model

7.1 Token Economics

Parameter	Value
Total Supply	84,000,000 LNC
Block Reward	Halving schedule
Block Time	~4 minutes
Consensus	Proof-of-Work (Script)

Table 7: Token Parameters

7.2 Fee Structure

Action	Fee
Standard TX	0.0001 LNC
UTXO Link Creation	0.001 LNC
Cross-Chain Swap	0.1%
DID Registration	0.01 LNC

Table 8: Fee Schedule

8 Roadmap

8.1 Phase 1: Foundation (Complete)

- ✓ IronTX Activation (Block 10,000)
- ✓ Schnorr Signatures
- ✓ Embedded Lightning
- ✓ Qt Wallet Integration

8.2 Phase 2: Core LinkNet (2026)

Quarter	Milestone	Component
Q2	Channel Factories	Link Channels
Q2	Atomic Swap CLI	Cross-Chain
Q3	UTXO Link Registry	UTXO Links
Q3	Data Link Indexer	Data Links
Q4	Virtual Channels	Link Channels
Q4	DID Method	Data Links

Table 9: 2026 Roadmap

8.3 Phase 3-4: Ecosystem & Applications (2027)

- Multi-party Channels
- Multi-chain Swaps
- LinkDEX Launch
- Mobile Wallet
- Identity Platform
- Enterprise API

9 Governance

9.1 Upgrade Mechanism

Change Type	Activation	Threshold
Soft Fork	Block height	95% miner signaling
Hard Fork	Block height	Community consensus
Protocol Extension	No fork	Opt-in adoption

Table 10: Upgrade Mechanisms

10 Conclusion

10.1 The Transformation

Linkcoin has evolved from an experimental memecoin fork into purpose-built infrastructure for the connected blockchain economy. The name “Linkcoin” now reflects its core mission: **linking** the fragmented blockchain landscape.

10.2 Key Differentiators

10.3 The Path Forward

With IronTX active and embedded Lightning operational, Linkcoin is positioned to deliver on the LinkNet vision:

1. **UTXO Links:** Programmable money without smart contract complexity
2. **Link Channels:** Scalable payments for the multi-party economy

Feature	Linkcoin	Bitcoin	Ethereum
Malleability Fix	IronTX	SegWit (partial)	N/A
Payment Channels	Embedded	External	L2s
Cross-Chain	Adaptor sigs	HTLC only	Bridges
Multi-party Channels	Link Channels	Limited	Contracts
Identity	Native DID	None	ENS

Table 11: Platform Comparison

3. **Cross-Chain Links:** Trustless bridges eliminating intermediaries
4. **Data Links:** Identity and attestation infrastructure

“Linking the future of value, data, and identity.”

A Technical Specifications

A.1 Network Parameters

Parameter	Mainnet	Testnet
P2P Port	7200	72555
RPC Port	7200	72555
Address Prefix	L	l
IronTX Height	12,000	12,000

A.2 Proposed Opcodes

Opcode	Hex	Purpose
OP_CHECKLNCEUTXO	0xc0	Verify linked UTXO in same TX
OP_CHECKUTXOSPENT	0xc1	Verify UTXO has been spent
OP_CHECKUTXOSPENTBY	0xc2	Verify UTXO spent by pubkey

B Glossary

Adaptor Signature Signature locked to external secret revelation

Channel Factory Multi-party funding for multiple channels

DID Decentralized Identifier (W3C standard)

IronTX Linkcoin's non-malleable transaction format

IronTxId Stable transaction ID excluding signatures

LinkNet Unified protocol layer on Linkcoin

MuSig2 Multi-signature aggregation scheme

Virtual Channel Channel through intermediary without trust

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